

**QA Report for
G-DF-LOGIC_MIXED_MODE13-1P8M-MMC_AL_
L130E_SHRINKABLE-CALIBRE-DRC
Ver.2.5_P1**

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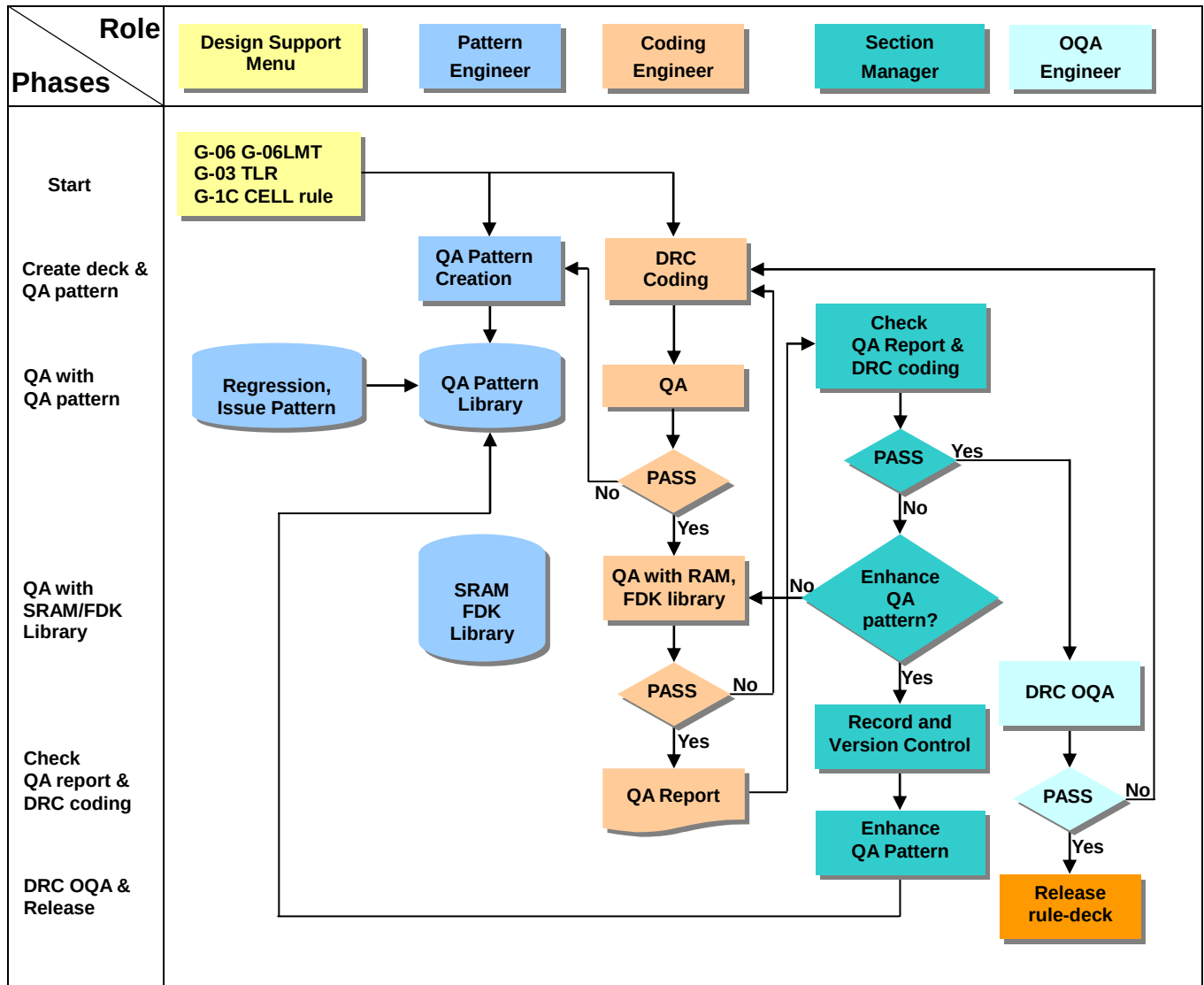
Preface

The purpose of this report is to describe the QA results of acceptance tests. This report will explain the various acceptance tests run on the DRC command file, and will cover the Layout Rules Check with QA Patterns, the Metal Option Check and the SRAM / Library Check.

Chapter 1. EDA Tool Used for Verifying this Deck

Classification	EDA Tool	Version
DRC Tool	Mentor Calibre	v2011.4_22.20

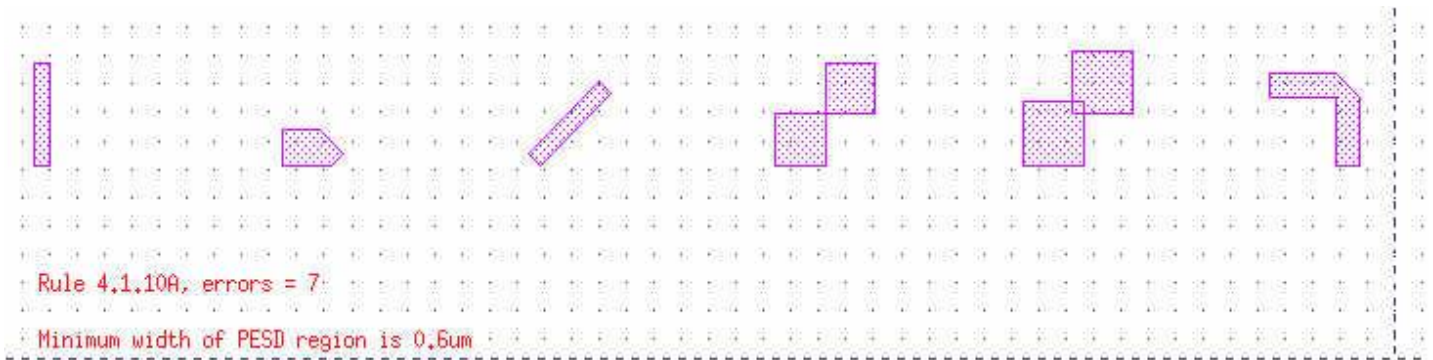
Chapter 2. DRC Deck Development Flow



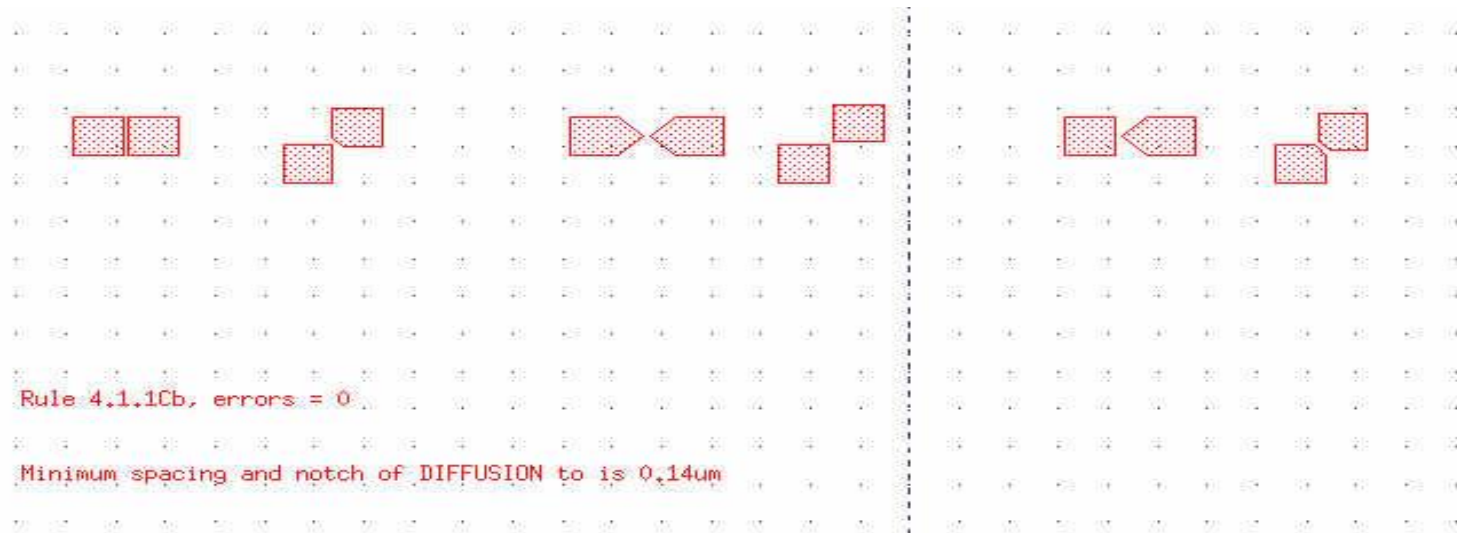
Chapter 3. Introduction to QA Patterns

The QA Patterns, designed for the DRC deck validation, include the fail cell and the pass cell and several kinds of test pattern for each rule.

Patterns for minimum width checking (fail cell)



Patterns for minimum spacing checking (pass cell)



Chapter 4. Layout Rules Check

The list of DRC rules in this DRC deck has been compared and made sure that it is consistent with that in UMC's TLR.

4-1. Check Result

QA Content	GDS File Name	Layout Cell	Purpose	Result
SAB rules	G_QA_SAB_C1_070912.gds	top	Detail QA	OK
DSR rules	L13_Die_Seal_Ring_100719.db	L13_DSR_POLYGON	Detail QA	OK
DSR rules	top_sealring_8K_051103.gds	pass_sealring_8K, fail_sealring_8K	Detail QA	OK
SLOT rules	G_QA_LOGIC_MIXED13_AL_METAL_SLOT_100115.gds	pass_SLOT, fail_SLOT	Detail QA	OK
DSR corner rules	top_die_sealring_corner_051107.gds	pass_die_sealring_corner, fail_die_sealring_corner	Detail QA	OK
Die corner rules	G_QA_015_below_die_sealring_corner_D_110712.gds	fail_die_sealring_corner, fail_die_sealring_corner_nsr, fail_die_sealring_corner_s, fail_die_sealring_corner_s_nsr, pass_die_sealring_corner, pass_die_sealring_corner_nsr, pass_die_sealring_corner_s, pass_die_sealring_corner_s_nsr	Detail QA	OK
logic rules	G_QA_LOGIC_MODE13_1P8M_MMC_AL_SHRINKABLE_V0.3_P1_G_111228.gds	TOPCELL	Detail QA	OK
Antenna rules	G_QA_LOGIC_MIXED13_1P8M_MMC_AL_L130E_SHRINKABLE_V2.3_P1_Topcell_200316.gds	Fail, Pass	Detail QA	OK

Chapter 5. Metal Option Check

5-1. Check Result

Metal Option File Name	Result
G-DF-LOGIC_MIXED_MODE13-1P4M1T-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P5M1T-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P5M2T-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P6M1T-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P6M2T-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P7M1T-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P7M2T-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P8M2T-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P4M1T_12KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P5M1T_12KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P5M2T_12KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P6M1T_12KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P6M2T_12KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P7M1T_12KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P7M2T_12KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P8M2T_12KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P4M1T_20KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P5M1T_20KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P5M2T_20KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P6M1T_20KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P6M2T_20KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P7M1T_20KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P7M2T_20KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P8M2T_20KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P4M1T_40KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P5M1T_40KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P5M2T_40KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P6M1T_40KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P6M2T_40KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P7M1T_40KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P7M2T_40KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK
G-DF-LOGIC_MIXED_MODE13-1P8M2T_40KA-MMC_AL_L130E_SHRINKABLE-CALIBRE-DRC-2.5_P1	OK

Chapter 6. SRAM and Library Check

6-1. Check Result

SRAM

Marker Layer Name	GDS File Name	Result
SPS	hs228_miniaarray_100_20090625a.db	OK
SPLLS	ll228_miniaarray_100_20090625a.db	OK
DP4S	L130E_8T_HSSP339_miniaarray_TB_070522.gds	OK
DP3S	L110E_8T_LL339_miniaarray_TC_123107.gds	OK
SP_4	L110_164LL_110124.db	OK
SP_5	L110_164SP_110124.db	OK
SP_16	141SPSRAM_MINIARRAY_20120109.db	OK
SP_18	141LLSRAM_MINIARRAY_20120109.db	OK